higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

INDUSTRIAL INSTRUMENTS N5

(8080205)

19 April 2023 (X-paper)
09:00–12:00

Drawing instruments and nonprogrammable calculators may be used.

This question paper consists of 6 pages and 1 formula sheet.

299Q1A2319

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
INDUSTRIAL INSTRUMENTS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a new page.
5. Use only a black or blue pen.
6. Write neatly and legibly.

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SECTION A

QUESTION 1

1.1 Explain each of the following terms:

1.1.1 Potential energy

1.1.2 Relative humidity

1.1.3 Gain

1.1.4 Polarisation

1.1.5 Stoke's law

(5 × 2) (10)

1.2 Give ONE term for each of the following descriptions by writing it next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

1.2.1 Point where flow tempo is at its maximum and pressure at its minimum

1.2.2 Circular metal plate with a central hole with the circumference of the plate

1.2.3 Vertical tube with a conical shape, the area gradually expanding from the bottom to the top and in it a disc free to move

1.2.4 Temperature at which saturation vapour pressure equals partial pressure of water vapour in the atmosphere

1.2.5 When fluids are deformed by strain, the ratio between shear rate and shear stress will not be a constant value.

(5 × 2) (10)

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1.3 Indicate whether the following statements are TRUE or FALSE by writing only 'True' or 'False' next to the question number (1.3.1–1.3.5) in the ANSWER BOOK.

1.3.1 As incompressible fluid moves through a restriction, velocity increases and pressure decreases.

1.3.2 Absolute viscosity is calculated by multiplying kinematic viscosity by fluid density.

1.3.3 A manipulated variable is the value with which a controller varies in its efforts to maintain the controlled variable at set point.

1.3.4 A turbine flowmeter does not require signal compensation for square compensation.

1.3.5 Bernoulli's equation is a mathematical expression of fluid density and compressibility in a restriction.

(5 × 1) (5)
[25]

TOTAL SECTION A: 25

SECTION B

QUESTION 2: FLOW MEASEREMENTS

2.1 Prove, by using Bernoulli's theorem for liquids, that the velocity in a pipe, when measured by means of a pitot tube, is equal to $V = c\sqrt{2gh}$. (6)

2.2 What is the basic difference between a *pitot tube* and an *annubar*? (2)

2.3 Define Faraday's law of electromagnetic induction. (3)

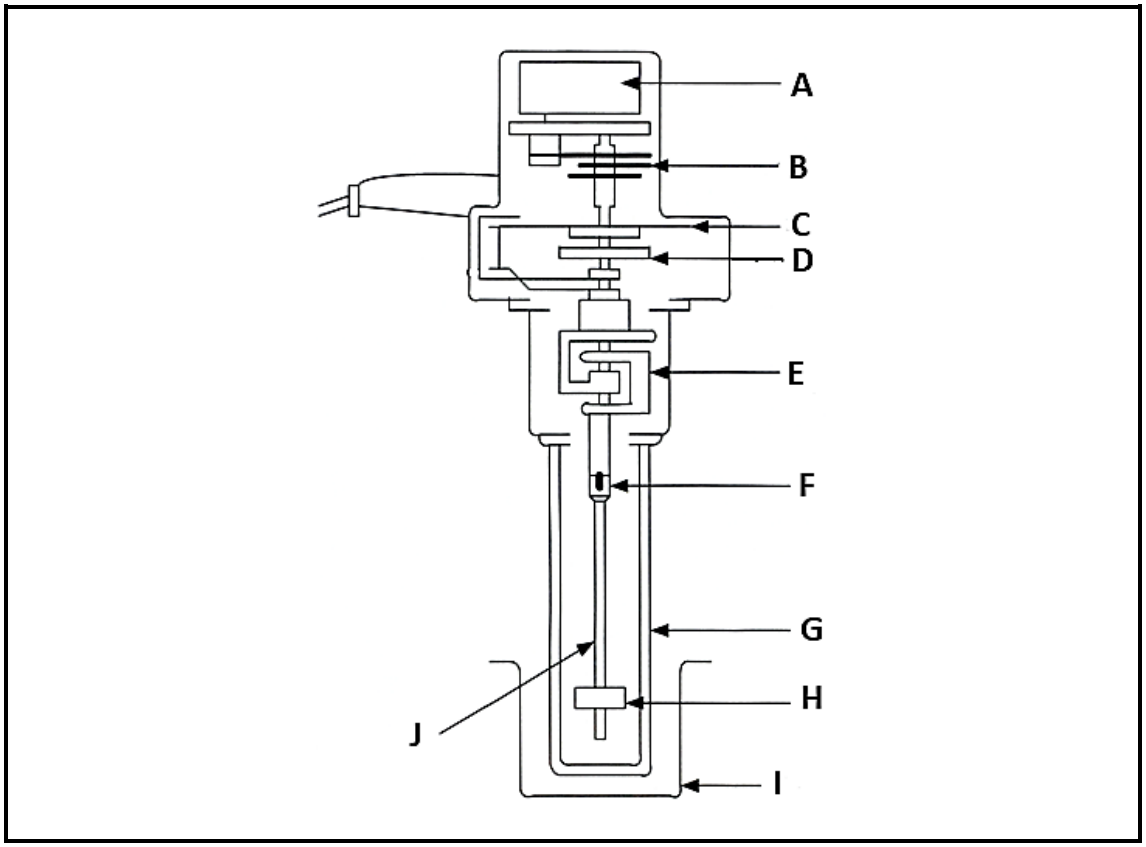
2.4 Make a neat, labelled sketch of a vortex-shedding flowmeter. (4)

2.5 Give THREE advantages and TWO disadvantages a vortex-shedding flowmeter. (3 + 2) (5)
[20]

QUESTION 3: DENSITY, HUMIDITY AND VISCOSITY

3.1 Make a neat, labelled sketch of a thermal-electrical dew point hygrometer. (7)

3.2



3.2.1 Identify the components of the viscometer by writing only the answer next to the letter (A–J) in the ANSWER BOOK. (10 × ½) (5)

3.2.2 Which type of viscometer is shown in the figure? (2)

3.2.3 State FIVE conditions to keep in mind when using the viscometer named in QUESTION 3.2.2. (5)

3.3 Explain, with the aid of a sketch, the operation of a sling-type hygrometer. (6)
[25]

QUESTION 4: pH MEASUREMENT

4.1 Calculate the $[H^+]$ and the $[OH^-]$ if the pH of a solution is 4,4. (6)

4.2 Name SIX types of measuring electrodes used with a reference electrode. (6)

4.3 Name THREE factors affecting measurement with potential-measuring instruments. (3)
[15]

QUESTION 5: AUTOMATIC CONTROL

5.1 Make a neat, labelled sketch of a three-term electronic controller using one operational amplifier. (7)

5.2 Explain the term *controller sensitivity*. (3)

5.3 What is the purpose of integral action? (1)

5.4 Explain the term *inherent control*. (1)

5.5 Explain the term *open-loop control*. (3)

[15]

TOTALSECTION B: 75
GRAND TOTAL: 100

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FORMULA SHEET

$$Q = \frac{8}{15} \tan\frac{\theta}{2}\sqrt{2g.H^5}$$

$$Q = \frac{2}{3}\ \mathrm{BC}\sqrt{2g.H^3}$$

$$Q = \frac{2}{3}\mathrm{CB}\sqrt{2g.H^3}$$

pH = -log[H+]

pOH = -log[OH-]

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